

Notes: Malamud, Schrimpf & Zhang (RFS, 2025)

An Intermediation-Based Model of Exchange Rates

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1 Big picture

- **Question.** Can intermediary market power in international financial markets explain exchange-rate puzzles such as deviations from uncovered interest parity (UIP), covered interest parity (CIP), and the carry trade premium?
- **Main idea.** International asset trading is intermediated. Households can directly trade domestic assets, but foreign claims must be accessed through global intermediaries in dealer-to-customer (D2C) markets.
- **Market structure.** Intermediaries have access to a frictionless dealer-to-dealer (D2D) market, but households do not. Intermediaries use bargaining power to charge state-contingent markups when households want exposure to foreign claims.
- **Core mechanism.** Real demand for goods creates demand for international risk-sharing. Intermediaries price discriminate across households and states. Their markups tilt household state prices and make international risk-sharing inefficient.
- **Why this matters for exchange rates.** Because exchange rates are tied to relative stochastic discount factors, intermediary markups distort risk premia, short rates, and exchange-rate dynamics even when intermediaries have zero net positions.
- **Theoretical result.** The paper derives closed-form equilibrium quantities in a continuous-time two-country Lucas-style cash-in-advance economy with endogenous intermediation markups.
- **Monetary neutrality result.** In frictionless markets, money is neutral. With intermediation frictions, markups break neutrality because monetary shocks affect the risk premia and exchange rates faced by households.
- **UIP result.** The model generates a positive Fama regression coefficient because intermediaries' price discrimination creates a negative relation between the interest-rate differential and the foreign-exchange risk premium.
- **Carry result.** A model with risk-neutral intermediaries and zero net positions can still generate a nontrivial carry premium because intermediary markups change the relative prices of currency risk.
- **CIP result.** CIP deviations arise endogenously from real demand pressures plus intermediary market power, not only from balance-sheet constraints.
- **Calibration result.** The frictional model matches several empirical patterns better than the frictionless benchmark: exchange-rate volatility, positive cross-country rate correlations, a large positive Fama beta, negative average CIP deviations, and a negative relation between CIP changes and exchange-rate changes.
- **Economic takeaway.** The paper turns FX puzzles into a market-structure problem: global intermediaries do not need to hold large inventories or face binding constraints to distort exchange rates; bargaining power and segmentation are enough.

2 Environment and measurement in the model

2.1 Two-country Lucas cash-in-advance economy

- There are two countries, Home H and Foreign F .
- Each country has a Lucas tree producing perishable goods, $X_{H,t}$ and $X_{F,t}$.
- Households in country i consume domestic goods $C_{i,t}$ and foreign goods $C_{i,t}^*$.
- Preferences are log-linear with home bias:

$$u_i(C_i, C_i^*) = \beta_i \log C_i + (1 - \beta_i) \log C_i^*, \quad i \in \{H, F\}.$$

- The preference shifter $\Psi_{i,t}$ is a demand or time-discount shock.

- Goods-market feasibility is

$$C_{H,t} + C_{F,t}^* = X_{H,t}, \quad C_{H,t}^* + C_{F,t} = X_{F,t}.$$

Interpretation: Home bias matters because international trade creates a motive to hold foreign claims. Without international goods trade, households would not need as much foreign-currency borrowing or insurance, and intermediary markups would have a weaker effect on FX prices.

2.2 Cash-in-advance and nominal output

- Country k goods must be bought with country k currency.
- The cash-in-advance condition is

$$P_{k,t} X_{k,t} = M_{k,t}, \quad k \in \{H, F\}.$$

- The nominal exchange rate E_t is the home-currency price of foreign currency. When E_t increases, foreign currency appreciates against home currency.
- Total domestic-currency spending by country- i households is

$$\bar{C}_{i,t} = C_{i,t} P_{i,t} + C_{i,t}^* P_{i,t}^*.$$

Interpretation: The model keeps the goods side close to the classic Lucas cash-in-advance setup. The novel part is not goods pricing; it is how international financial claims are intermediated.

2.3 Segmented financial markets

- Households can trade domestic bonds and domestic trees directly in all-to-all local markets.
- To trade foreign assets or state-contingent foreign claims, households must contact global intermediaries in the D2C market.
- Intermediaries can trade freely in a complete D2D market.
- The D2D pricing kernels satisfy the no-arbitrage relation

$$\frac{M_{F,T}^I}{M_{F,t}^I} = \frac{M_{H,T}^I}{M_{H,t}^I} \frac{E_T}{E_t}.$$

Interpretation: The wedge between household market access and intermediary market access is the source of endogenous market incompleteness. Households face distorted state prices even though intermediaries themselves can hedge in complete markets.

2.4 The dealer-to-customer bargaining problem

- Intermediaries know households' country of origin and can charge country-specific markups.
- A household's D2C kernel $M_{i,t,t+dt}$ can differ from the D2D kernel $M_{i,t,t+dt}^I$.
- Intermediaries choose the terms of access subject to household participation and no-arbitrage constraints for locally traded assets.
- The bargaining solution makes the markup security-specific: securities that provide valuable insurance against desired states can command larger markups.

Interpretation: The intermediation wedge is not a generic transaction cost. It is state-contingent and related to the household's desire for insurance. This is why it can distort risk premia and exchange rates.

2.5 Main state variables

The model uses two key state variables:

$$Q_t = \frac{M_{F,t} E_t}{M_{H,t}},$$

which is the exchange rate adjusted for relative money supplies, and

$$\pi_{i,t} = \frac{\beta_i \bar{C}_{i,t}}{M_{i,t}},$$

which is the nominal consumption share spent on domestic goods.

Interpretation: Q_t summarizes the endogenous redistribution of wealth across countries. As Q_t changes, the domestic goods shares $\pi_{H,t}$ and $\pi_{F,t}$ shift, changing demand for foreign claims and therefore the markups intermediaries can charge.

3 Models and key equations

3.1 Diffusion shocks and stochastic discount factors

The fundamentals are driven by a Brownian vector B_t :

$$\frac{d\Psi_{i,t}}{\Psi_{i,t}} = -\delta dt + (\theta_{i,t}^\Psi)' dB_t, \quad \frac{dM_{i,t}}{M_{i,t}} = \mu_i dt + \theta_i' dB_t.$$

All stochastic discount factors admit the diffusion representation

$$-\frac{dM_{i,t}}{M_{i,t}} = r_{i,t} dt + \eta_{i,t}' dB_t, \quad -\frac{dM_{i,t}^I}{M_{i,t}^I} = r_{i,t}^I dt + (\eta_{i,t}^I)' dB_t.$$

Interpretation: The vectors $\eta_{i,t}$ and $\eta_{i,t}^I$ are risk premia faced by households and intermediaries. The model's key object is the difference between them.

3.2 Proposition 1: markup-induced risk-premium wedge

The paper characterizes how the household pricing kernel differs from the interdealer pricing kernel. In the notation of the notes, the wedge can be summarized as

$$\eta_{i,t} - \eta_{i,t}^I = \text{function of household demand, local constraints, and intermediary markups.}$$

The strength of the wedge depends on the intermediation adjustment parameter and on how much the household wants securities that intermediaries can provide.

Interpretation: Intermediaries tilt state prices in the direction households most want. Insurance that is most valuable to a household is exactly where the intermediary can charge the greatest rent.

3.3 Lemma 2: exchange rates and consumption shares

With home bias $\beta_H > 1/2$ and $\beta_F > 1/2$, the domestic consumption shares are functions of the adjusted exchange rate:

$$\pi_{H,t} = \left(\frac{\beta_F}{1 - \beta_F} - Q_t \right) \left(\frac{\beta_F}{1 - \beta_F} - \frac{1 - \beta_H}{\beta_H} \right)^{-1},$$

$$\pi_{F,t} = \left(\frac{\beta_H}{1 - \beta_H} - \frac{1}{Q_t} \right) \left(\frac{\beta_H}{1 - \beta_H} - \frac{1 - \beta_F}{\beta_F} \right)^{-1}.$$

Interpretation: When Q_t moves, wealth and spending shares shift across countries. This creates a feedback loop: exchange rates change risk-sharing demand, and risk-sharing demand changes intermediation markups and exchange rates.

3.4 Proposition 2: frictionless equilibrium

Without intermediation markups, risk premia are

$$\eta_{i,t}^C = \underbrace{\theta_i - \theta_{i,t}^\Psi}_{\text{domestic risk}} + \underbrace{(1 - \pi_{i,t})(\theta_{i,t}^\Psi - \theta_{-i,t}^\Psi)}_{\text{risk sharing}},$$

while short rates are

$$r_{i,t}^C = \mu_i + \delta + \theta'_i \eta_{i,t}^C.$$

Interpretation: In frictionless markets, money is neutral. Monetary shocks are absorbed by prices and exchange rates, and only real hedging motives matter for risk premia.

3.5 Proposition 3: equilibrium risk premia with intermediation

When intermediation frictions are present, the interdealer risk premium has an extra distortion. In the zero-adjustment-cost case emphasized in the paper,

$$\eta_{i,t}^I = \underbrace{(2 - \lambda_{i,t})\theta_i - \theta_{i,t}^\Psi}_{\text{domestic risk}} + \underbrace{\frac{1 - \pi_{i,t}}{\pi_t} [\theta_{i,t}^\Psi - \theta_{-i,t}^\Psi + (\lambda_{i,t} - 1)\theta_i - (\lambda_{-i,t} - 1)\theta_{-i}]}_{\text{risk sharing}}.$$

Interpretation: Intermediaries change not only the price of foreign risk-sharing but also the way monetary shocks enter the risk premium. This is the precise sense in which market power breaks monetary neutrality.

3.6 Interest-rate effects of intermediation rents

The model decomposes equilibrium short rates into a frictionless term, intermediation rents, and a risk-premium alignment term:

$$\begin{aligned} r_{i,t} = & \delta + \mu_i - \|\eta_{i,t}^I\|^2 - (\eta_{i,t}^I)' [\pi_{i,t}\theta_{i,t}^\Psi + (1 - \pi_{i,t})\theta_{-i,t}^\Psi] \\ & + \pi_{i,t}I_{i,t} + (1 - \pi_{i,t})I_{-i,t} \\ & + (\eta_{i,t}^I)' [\pi_{i,t}(\eta_{i,t}^I - \eta_{i,t}) + (1 - \pi_{i,t})(\eta_{-i,t}^I - \eta_{-i,t})]. \end{aligned}$$

Interpretation: Intermediation rents push up equilibrium short rates. The result is surprising because intermediaries are risk-neutral and carry no net positions, yet their pricing power alters aggregate interest-rate dynamics.

3.7 Demand-shock volatility and the calibration state

The calibration introduces persistent volatility in demand shocks:

$$\|\theta_{i,t}^\Psi\|^2 = (\sigma_t^\Psi)^2, \quad \sigma_t^\Psi = \exp(4x_t^2) - 1,$$

where

$$dx_t = -\kappa_x x_t dt + \sigma_x \sqrt{1 - x_t^2} dB_t^x.$$

Interpretation: The model needs fluctuations in the magnitude of risk-sharing demand. The exogenous state x_t governs the intensity of demand shocks; the endogenous state Q_t governs trade and wealth redistribution.

3.8 Proposition 4: CIP deviations

The D2C borrowing rate for country- i households borrowing in country $-i$ currency equals the direct rate plus a markup. Under Assumption 2,

$$CIP_{i,t} = \underbrace{\frac{(1 - \pi_{i,t})(\pi_t - 1)}{2\pi_t^2}}_{\text{risk-sharing demand}} \|\theta_{i,t}^\Psi - \theta_{-i,t}^\Psi\|^2 < 0.$$

Interpretation: CIP deviations arise from the same household risk-sharing demand that generates UIP deviations. Foreign-currency borrowing becomes expensive because intermediaries charge markups for access to foreign claims.

3.9 Fama regression and carry trade returns

The Fama regression is

$$(r_{H,t} - r_{F,t})dt - \log E_{t+dt} + \log E_t = \alpha + \beta^{Fama}(r_{H,t} - r_{F,t})dt + \varepsilon_{t+dt}.$$

The carry trade return is

$$Carry_{t+dt} = \text{sign}(r_{H,t} - r_{F,t}) [(r_{H,t} - r_{F,t})dt - \log E_{t+dt} + \log E_t].$$

Interpretation: UIP predicts a very different relation between interest-rate differentials and exchange-rate changes than what is observed. In this model, intermediary price discrimination generates a risk premium that can reverse the usual UIP logic.

4 Conceptual design and calibration logic

4.1 What is disciplined by theory

- The model's equilibrium objects are solved in closed form.
- Households' direct access to domestic assets and indirect access to foreign claims generates market segmentation.
- Intermediaries bargain with households and set state-contingent markups.
- The D2D market clears intermediaries' exposures, so intermediaries can have zero net positions.
- Risk-sharing demand and goods trade jointly determine markups and exchange-rate dynamics.

Interpretation: The paper is not identifying a treatment effect from data. It is deriving a mechanism and then asking whether a calibrated version can match known moments.

4.2 What is disciplined by calibration

- Table 1 chooses parameters to target trade-to-GDP, output-growth volatility, consumption-growth volatility and correlation, exchange-rate volatility, and interest-rate-differential persistence.
- The calibration compares a frictionless model with a frictional model.
- The frictionless benchmark removes intermediation markups.
- The frictional model keeps the same underlying macro environment but adds segmented D2C markets and intermediary market power.

Interpretation: The relevant comparison is not between two arbitrary economies. It is between the same macro environment with and without intermediation markups.

4.3 What moments matter most

- Exchange-rate volatility and autocorrelation.
- Consumption-growth and output-growth comovement.
- Interest-rate-differential volatility and persistence.
- Cross-country correlations of interest rates and rate changes.
- Fama beta and carry trade Sharpe ratio.
- Mean CIP deviation and the relation between CIP changes and exchange-rate changes.

Interpretation: The calibration is strongest where intermediation has direct pricing implications: interest rates, UIP, carry returns, and CIP deviations.

4.4 What the paper cannot fully prove

- The model abstracts from binding dealer balance-sheet constraints to isolate market power.
- Some parameter choices, especially preference or demand shock processes, are difficult to measure directly.
- The symmetric benchmark cannot fully match empirical carry trade magnitudes.
- Some richer quantitative matches rely on Internet Appendix extensions with asymmetric country exposures.
- The model's equity premia and equity Sharpe ratios are too low in the baseline with log preferences.

Interpretation: The paper's main contribution is not that it perfectly explains every asset-pricing moment. The contribution is that intermediary market power alone can generate major exchange-rate puzzles in general equilibrium.

5 Figures, tables, and original equation panels

The paper has one main figure and two main calibration tables. Because it is heavily theoretical, I also include cropped original equation and proposition panels. Adjacent panels are grouped to save space and make comparison faster.

5.1 Market structure and baseline economy

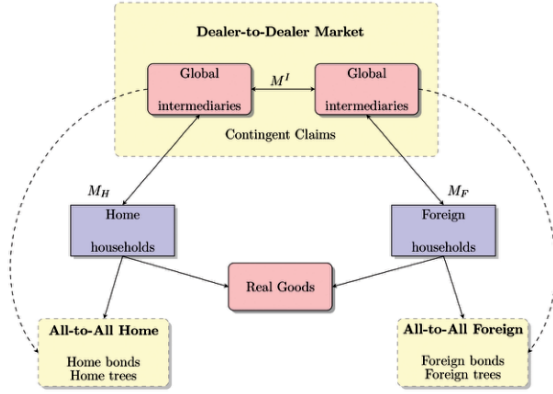


Figure 1
Graphical description of market structure in our model.

(a) Figure 1: market structure.

with a Lucas tree producing a strictly positive amount of country-specific perishable goods, $X_{H,t}$ and $X_{F,t}$, respectively, for $t \geq 0$. Before the financial markets open, trees are owned by the respective country's households.

Each country is populated by a continuum of identical households whose preferences are represented by a time-additive log-linear utility over the consumption of both goods,

$$E \left[\int_0^\infty \Psi_{i,t} u_i(C_{i,t}, C_{i,t}^*) dt \right], \quad (1)$$

with

$$u_i(C_i, C_i^*) = \beta_i \ln(C_i) + (1 - \beta_i) \ln(C_i^*), \quad i \in \{H, F\}.$$

Here, β_H, β_F are the weights on the Home goods in the utility of each country's households, accounting for the potential home bias in consumption. As is common in the literature, we use $*$ to denote goods and quantities of the foreign country. The time-preference "demand shocks" $\Psi_{H,t}, \Psi_{F,t}$ (henceforth, time discount factors) are arbitrary positive random variables, normalized so that $\Psi_{i,0} = 1$. Such demand shocks are commonly used in international economics. See, for example, [Stockman and Tesar \(1995\)](#), [Pavlova and Rigobon \(2007\)](#), and [Itskhoki and Mukhin \(2021\)](#).⁷ Equilibrium prices are then pinned down by imposing market clearing for all goods (the economy-wide feasibility constraint):

$$C_{H,t} + C_{F,t}^* = X_{H,t} \quad (\text{home goods market clearing})$$

(b) Utility and market-clearing setup.

Read:

- Figure 1 shows the key segmentation: households can trade domestic assets locally, but foreign contingent claims require intermediaries.
- Intermediaries trade among themselves in a centralized D2D market.
- The utility/market-clearing panel shows the two-country home-bias setup and goods-market feasibility conditions.

Economic message: The entire model turns on the wedge between household access and intermediary access. This wedge lets intermediaries price discriminate and distort state prices.

5.2 Pricing kernels and D2C bargaining

$$\|\theta\|^2 = \sum_{k=1}^N \theta_k^2 \quad (12)$$

to denote the squared Euclidean norm of a vector $\theta \in \mathbb{R}^N$.

We assume that the fundamentals of the economy, $(\Psi_{H,t}, \Psi_{F,t}, \mathcal{M}_{H,t}, \mathcal{M}_{F,t})$, are driven by an N -dimensional standard Brownian motion B_t . The time preference shocks follow

$$\frac{d\Psi_{i,t}}{\Psi_{i,t}} = -\delta dt + (\theta_{i,t}^\Psi)' d B_t,$$

while nominal output in the two countries follows

$$\frac{d\mathcal{M}_{i,t}}{\mathcal{M}_{i,t}} = \mu_i dt + \theta_{i,t}' d B_t, \quad i \in \{H, F\},$$

with some shock exposure vectors $\theta_i, \theta_{i,t}^\Psi \in \mathbb{R}^N$. In particular, the volatility of the nominal output is given by

$$\text{Var}\left[\frac{d\mathcal{M}_{i,t}}{\mathcal{M}_{i,t}}\right] = \|\theta_i\|^2 dt,$$

where $\|\theta_i\|^2$ is defined in (12). We purposely assume that the time discount rates, $\Psi_{i,t}$, have the same drift, ensuring that, on average, households in the two countries are equally patient. This assumption is necessary to ensure the existence of a well-defined ergodic distribution of the model, in which both agents survive. For simplicity, we assume that all parameters of the diffusion processes above are constant, except for the time preference shocks $\theta_{i,t}^\Psi$. The randomness of $\theta_{i,t}^\Psi$ will play an important role in our calibration exercise. However, all our closed-form solutions hold for arbitrary stochastic dynamics of all coefficients. Since $\Psi_{i,t}$ are geometric Brownian motions, formula (10) simplifies and we have $D_{i,t} = \Psi_{i,t}/\delta$.

The continuous time assumption makes the optimal contracting problem in the D2C market particularly tractable. Indeed, standard arguments imply that all stochastic discount factors in the diffusion setting admit a representation

$$-\frac{d\mathcal{M}_{i,t}}{\mathcal{M}_{i,t}} = r_{i,t} dt + (\eta_{i,t})' d B_t, \quad -\frac{d\Psi_{i,t}^I}{\Psi_{i,t}^I} = r_{i,t}^I dt + (\eta_{i,t}^I)' d B_t, \quad (13)$$

where $r_{i,t}$ and $r_{i,t}^I$ are the risk-free rates in the two market segments, and

(a) Diffusion shocks and SDF representation.

in the D2C market this is coherent with what holds in the D2C market, and the same is true for the risk premia of the trees (condition (8)). Formally, conditions (7)–(8) can be rewritten as¹⁰

$$E_t\left[\frac{d\mathcal{M}_{i,t}}{\mathcal{M}_{i,t}}\right] = E_t\left[\frac{d\Psi_{i,t}^I}{\Psi_{i,t}^I}\right] \Leftrightarrow r_{i,t} = r_{i,t}^I \quad (14)$$

and

$$E_t\left[\frac{d\mathcal{M}_{i,t}}{\mathcal{M}_{i,t}} \frac{d\mathcal{M}_{i,t}}{\mathcal{M}_{i,t}}\right] = E_t\left[\frac{d\Psi_{i,t}^I}{\Psi_{i,t}^I} \frac{d\Psi_{i,t}^I}{\Psi_{i,t}^I}\right]. \quad (15)$$

Similarly, the objective (9) can be rewritten as

$$\max_{\eta_{i,t}} \left(E_t\left[\frac{d\mathcal{M}_{i,t}}{\mathcal{M}_{i,t}} \frac{dW_{i,t}}{W_{i,t}}\right] - E_t\left[\frac{d\Psi_{i,t}^I}{\Psi_{i,t}^I} \frac{dW_{i,t}}{W_{i,t}}\right] + E_t\left[\frac{d\mathcal{M}_{i,t}}{\mathcal{M}_{i,t}}\right] - E_t\left[\frac{d\Psi_{i,t}^I}{\Psi_{i,t}^I}\right] \right) \quad (16)$$

under the constraints (14)–(15). Applying Ito's lemma to $W_{i,t} = \delta^{-1} \Psi_{i,t} \mathcal{M}_{i,t}^{-1}$, we obtain the following expression for household wealth dynamics

$$\frac{dW_{i,t}}{W_{i,t}} = \underbrace{\frac{d\Psi_{i,t}}{\Psi_{i,t}}}_{\text{time discount}} - \underbrace{\frac{d\mathcal{M}_{i,t}}{\mathcal{M}_{i,t}}}_{\text{interest + premium}} + \underbrace{\left(\frac{d\mathcal{M}_{i,t}}{\mathcal{M}_{i,t}}\right)^2}_{\text{convexity}} - \underbrace{\frac{d\Psi_{i,t}}{\Psi_{i,t}} \frac{d\mathcal{M}_{i,t}}{\mathcal{M}_{i,t}}}_{\text{comovement}}. \quad (17)$$

(b) Dealer problem and wealth condition.

Read:

- The SDF panel shows how the model maps money and demand shocks into short rates and risk premia.
- The dealer panel shows the bargaining problem that links the D2C kernel to household wealth and intermediary markups.
- The key object is the difference between $\eta_{i,t}$ and $\eta_{i,t}^I$.

Economic message: Intermediation frictions are not appended as exogenous spreads. They arise from a bargaining problem over state-contingent access to foreign financial claims.

5.3 Proposition 1 and Lemma 2

Proposition 1. The optimal, markup-maximizing vector of risk premia $\eta_{i,t}$ chosen by the intermediary for country- i customers is given by

$$\eta_{i,t} = \frac{\Gamma}{2+\Gamma} \eta'_{i,t} + \frac{2}{2+\Gamma} \left[\frac{1}{2} (\eta'_{i,t} - \theta'_{i,t} \Psi + \lambda_{i,t} \theta_t) \right], \text{ with } \lambda_{i,t} = \frac{(\eta'_{i,t} + \theta'_{i,t} \Psi) \theta_t}{\|\theta_t\|^2}. \quad (21)$$

Proposition 1 shows explicitly how the strength of the intermediation frictions depends on the adjustment cost parameter Γ . When Γ is large, the financier faces a high adjustment cost and, as a consequence, decides to equalize the quoted SDF to that in the D2D market: $\eta_{i,t} = \eta'_{i,t}$. This corresponds to the equilibrium in a model without intermediation frictions. By contrast, when $\Gamma=0$, the financier faces no adjustment cost, and the intermediary selects the pricing kernel that maximizes rents.

Effectively, $\eta'_{i,t}$ represents the shadow costs of holding risk for intermediaries. Intermediaries pass this cost through to households, and (21) implies that the pass-through coefficient is given by

$$\frac{\partial \eta_{i,t}}{\partial \eta'_{i,t}} = \frac{1+\Gamma}{2+\Gamma}. \quad (22)$$

(a) Proposition 1: markup-driven risk-premium wedge.

We now introduce two key state variables that will serve as the endogenous Markov state driving the equilibrium dynamics. We let $\pi_{i,t} = \frac{\beta_i \bar{C}_{i,t}}{\mathcal{M}_{i,t}}$, $i=H,F$ denote the nominal consumption share for the domestic goods, and arrive at the following result.

Lemma 2. (Equilibrium Link Between Consumption Shares and Exchange Rates). Let $\mathcal{Q}_t = \frac{\mathcal{M}_{F,t} \bar{E}_t}{\mathcal{M}_{H,t}}$ be the value of one unit of home goods in terms of foreign goods, adjusted for money supply. Then,

$$\pi_{H,t} = \left(\frac{\beta_F}{1-\beta_F} - \mathcal{Q}_t \right) \left(\frac{\beta_F}{1-\beta_F} - \frac{1-\beta_H}{\beta_H} \right)^{-1},$$

$$\pi_{F,t} = \left(\frac{\beta_H}{1-\beta_H} - \frac{1}{\mathcal{Q}_t} \right) \left(\frac{\beta_H}{1-\beta_H} - \frac{1-\beta_F}{\beta_F} \right)^{-1},$$

and the adjusted exchange rate satisfies

$$\mathcal{Q}_t \in \left(\frac{1-\beta_H}{\beta_H}, \frac{\beta_F}{1-\beta_F} \right). \quad (25)$$

Lemma 2 shows how fluctuations of the adjusted exchange rate inside the (25) interval lead to a redistribution of wealth between the two countries. When $\mathcal{Q}_t$ appreciates, F -households become richer, $\pi_{F,t}$ increases, and $\pi_{H,t}$ decreases.

(b) Lemma 2: consumption shares and adjusted exchange rate.

Read:

- Proposition 1 characterizes how markups alter the household-facing pricing kernel.
- The strength of the intermediation wedge depends on the adjustment-cost parameter and the household's desired exposure.
- Lemma 2 links the adjusted exchange rate $\mathcal{Q}_t$ to domestic consumption shares $\pi_{H,t}$ and $\pi_{F,t}$.

Economic message: The model's feedback loop is visible here: exchange rates determine consumption shares, consumption shares determine demand for foreign assets, and that demand determines intermediation markups.

5.4 Frictionless versus frictional risk premia

use $X^\sim$ to denote a quantity X in the case of complete (frictionless) markets.

Proposition 2 (Frictionless Equilibrium). Without intermediation markups, the risk premia are given by

$$\eta_{i,t}^C = \underbrace{\theta_t - \theta'_{i,t} \Psi}_{\text{domestic risk}} + \underbrace{(1 - \pi_{i,t})(\theta'_{i,t} \Psi - \theta'_{-i,t} \Psi)}_{\text{risk sharing}}, \quad (26)$$

while the equilibrium interest rates are given by

$$r_{i,t}^C = \mu_i + \delta + \theta'_i \eta_{i,t}^C. \quad (27)$$

Absent intermediation frictions, money is neutral, and, hence, adjustments in exchange rates fully undo any effect of the money supply. No sharing of risks arising from monetary policy shocks is possible (and neither is it necessary since money is neutral). Nominal risk premia in the country i only depend on domestic shocks plus the amount of trade (captured by the share of foreign assets $1 - \pi_{i,t}$ times the risk sharing needs as captured by the difference

(a) Proposition 2: frictionless equilibrium.

Proposition 3 (Equilibrium Risk Premia with Intermediation). In equilibrium, risk premia are given by

$$\begin{aligned} \frac{1+\Gamma}{2+\Gamma} \eta'_{i,t} = & \theta_t - \left(\frac{1+\Gamma}{2+\Gamma} \theta'_{i,t} \Psi + \frac{1}{2+\Gamma} \lambda_{i,t} \theta_t \right) \\ & - (\theta_{-i} - \theta_t) \left(-1 + \frac{\pi_{-i,t} \alpha_t}{(1+\alpha_t)(2-2\pi_t)} + \frac{(1-\pi_{-i,t})}{(2\pi_t-1)(1+\alpha_t)} \right) \\ & - [\theta'_{i,t} \Psi - \theta'_{-i,t} \Psi] \left(\pi_{-i,t} \left(\frac{1}{1+\alpha_t} \frac{1}{2+\Gamma} - 1 \right) + \frac{\alpha_t}{1+\alpha_t} \right) \\ & - [\lambda_{-i,t} \theta_{-i} - \lambda_{i,t} \theta_t] \left(\pi_{-i,t} \frac{1}{1+\alpha_t} \frac{1}{2+\Gamma} - (1-\pi_{-i,t}) \frac{1}{1+\alpha_t} \frac{\alpha_t}{1+\alpha_t} \right). \end{aligned}$$

In the case when $\Gamma=0$, this expression simplifies to

$$\eta'_{i,t} = \underbrace{(2 - \lambda_{i,t}) \theta_t - \theta'_{i,t} \Psi}_{\text{domestic risk}} + \underbrace{\frac{1 - \pi_{i,t}}{\pi_t} (\theta'_{i,t} \Psi - \theta'_{-i,t} \Psi + (\lambda_{i,t} - 1) \theta_t - (\lambda_{-i,t} - 1) \theta_{-i})}_{\text{risk sharing}},$$

(b) Proposition 3: risk premia with intermediation.

Read:

- Proposition 2 decomposes frictionless risk premia into domestic risk and international risk sharing.
- Proposition 3 adds the intermediation wedge and the Lagrange multiplier terms.

- With intermediation, monetary shocks enter equilibrium risk premia and exchange rates.

Economic message: The comparison isolates the paper’s central theoretical result: market power breaks monetary neutrality and changes the mapping from shocks to FX risk premia.

5.5 Interest-rate dynamics and demand-shock volatility

rates are pinned down as the quotient of intermediaries’ pricing kernels. The dynamics of these kernels follow (13), and an application of the Ito formula leads to

$$d \ln \mathcal{E}_t = \underbrace{(r_{H,t}^I - r_{F,t}^I)}_{UIP} + \underbrace{\frac{1}{2}(\|\eta_{H,t}^I\|^2 - \|\eta_{F,t}^I\|^2)}_{\text{Currency risk premium}} dt + (\eta_{H,t}^I - \eta_{F,t}^I)' dB_t, \quad (30)$$

where the risk premium differentials are given by

$$\eta_{H,t}^I - \eta_{F,t}^I = \theta_H - \theta_F - \frac{1-\pi_t}{\pi_t}(\theta_{H,t}^\Psi - \theta_{F,t}^\Psi) + (\lambda_{H,t} - 1)\theta_H - (\lambda_{F,t} - 1)\theta_F.$$

When the cross-market (between D2D and D2C segments) arbitrage constraints in (20) bind, so that the Lagrange multiplier $\lambda_{i,t} - 1 \neq 0$, intermediation frictions create exposure of risk premia to monetary shocks, θ_t . When the Lagrange multipliers are low, households have no incentives to share monetary shocks. As a result, these shocks do not affect exchange rates. In equilibrium, the degree of monetary pass-through is determined by the degree of international trade. This stochastic pass-through of monetary shocks to exchange rates is a unique implication of our model.

As we show in the Appendix (see the proof of Proposition 2), equilibrium interest rates are given by

$$\begin{aligned} r_{i,t} = & \underbrace{\delta + \mu_i - \|\eta_{i,t}^I\|^2 - (\eta_{i,t}^I)'(\pi_{i,t}\theta_{i,t}^\Psi + (1-\pi_{i,t})\theta_{-i,t}^\Psi)}_{r_{i,t}^C} \\ & + \underbrace{\pi_{i,t}\mathcal{I}_{i,t} + (1-\pi_{i,t})\mathcal{I}_{-i,t}}_{\text{global rents}} \\ & + (\eta_{i,t}^I)'(\pi_{i,t}(\eta_{i,t}^I - \eta_{i,t}) + (1-\pi_{i,t})(\eta_{-i,t}^I - \eta_{-i,t})). \end{aligned} \quad (31)$$

(a) Exchange-rate and short-rate dynamics.

in the sequel, we focus our discussion on the zero-adjustment cost case ($\iota = 0$). Then, under Assumption 2, short-term rates follow

$$\begin{aligned} r_{i,t} = & \underbrace{\mu_i + \delta - \|\theta_t\|^2}_{r_i^C} \\ & + \underbrace{\frac{1-\pi_{i,t}(1-\pi_{-i,t}) + (\pi_{i,t} - \pi_{-i,t})(2\pi_t - 2)}{2\pi_t} \|\theta_{i,t}^\Psi - \theta_{-i,t}^\Psi\|^2}_{\text{intermediation frictions}}, \end{aligned} \quad (32)$$

where r_i^C is the short rate in the frictionless model (27). By direct calculation, the correction to the short-term rate (the second line in (32)) is always positive, consistent with our discussion of the formula (31).

Since r_i^C is constant, intermediation frictions generate volatility in the short rates (32) through fluctuations in international trade, as captured by π_t and $\pi_{i,t}$, as well as demand shocks, as captured by $\theta_{i,t}^\Psi$. In both the data and realistic calibrations, π_t is persistent and slow-moving. As a result, generating realistic interest rate volatility requires an additional source of fluctuations in $\theta_{i,t}^\Psi$. We model this source as time-varying, persistent, and mean-reverting volatility of demand shocks (see, e.g., Dahlquist et al. 2023).

Assumption 3 (Volatility of the Demand Shocks). We have

$$\|\theta_{i,t}^\Psi\|^2 = (\sigma_i^\Psi)^2, \quad i = H, F,$$

where $\sigma_i^\Psi = \exp(4x_t^2) - 1$, with x_t being a standard Jacobi process $x_t \in [-1, 1]$,

(b) Short-rate correction and demand-shock volatility.

Read:

- The interest-rate formula decomposes short rates into a frictionless component plus intermediation rents and risk-premium alignment terms.
- Intermediation rents push global short rates up.
- The demand-shock volatility state x_t generates fluctuations in risk-sharing demand and interest-rate differentials.

Economic message: Interest-rate volatility in the model comes from endogenous intermediation rents responding to demand for risk sharing, not from exogenous dealer balance-sheet constraints.

5.6 CIP deviations and calibration parameters

simple, closed-form expression for these deviations.

Proposition 4 (CIP Deviations). The short rate for borrowing in the country

($-i$) currency quoted D2C to the country i household, $r_{i,t}^{-i}$, is given by

$$\begin{aligned} r_{H,t}^F &= \underbrace{r_{F,t}}_{\text{direct country-F rate}} + \underbrace{(\eta_{H,t}^I - \eta_{F,t}^I)(\eta_{H,t}^I - \eta_{H,t}^I)}_{\text{markup}} \\ r_{F,t}^H &= \underbrace{r_{H,t}}_{\text{direct country-H rate}} + \underbrace{(\eta_{F,t}^I - \eta_{H,t}^I)(\eta_{F,t}^I - \eta_{F,t}^I)}_{\text{markup}}. \end{aligned}$$

Under [Assumption 2](#), we have

$$CIP_{i,t} = \frac{(1 - \pi_{i,t})(\pi_t - 1)}{2(\pi_t)^2} \underbrace{\|\theta_{i,t}^\Psi - \theta_{-i,t}^\Psi\|^2}_{\text{risk sharing}} < 0, \quad i = H, F.$$

(a) Proposition 4: CIP deviations.

Table 1

Parameter choices for the simulated moments

Variable Definitions	Symbols	Values	Targeted Moments
Preferences	$\{\beta_i\}_{i=H,F}$	0.9	Trade-to-GDP ratio 0.2
Time discount	δ	0.03	—
Drift of $\mathcal{M}_{i,t}$	μ	0.03	—
Size of Supply shocks	$\bar{\sigma}^{\mathcal{M}}$	0.014	$\text{std}(dg_H) = 2\%$
Supply shocks correlation	$\rho^{\mathcal{M}}$	0.7	$\text{corr}(dg_H, dg_F) = 0.35$
Correlated demand shocks	σ_{Ψ}^{Ψ}	$\exp(4x_t^2) - 1$	—
Demand shocks correlation	ρ^{Ψ}	0.3	$\text{std}(de) = 10\%$
Idiosyncratic demand shocks	$\bar{\sigma}^{\Psi}$	0.095	$\text{corr}(dc_H, dc_F) = 0.3$
Output-demand correlation	s^{Ψ}	-0.4	$\text{corr}(dc_H, dc_F) = 0.3$
Mean-reversion of x_t	κ^x	0.36/12	$\text{ac1}(r_H - r_F) = 0.95$
Volatility parameter of x_t	σ^x	0.09/2	$\text{std}(r_H - r_F) = 0.6\%$

(b) Table 1: parameter choices.

Read:

- Proposition 4 shows that D2C foreign-currency borrowing rates equal direct rates plus markups.
- Under the paper's assumptions, $CIP_{i,t} < 0$: borrowing foreign currency through the household-facing market is more expensive.
- Table 1 sets home-bias at $\beta_i = 0.9$, discount rate at $\delta = 0.03$, supply shock volatility at $\bar{\sigma}_M = 0.014$, and demand-shock volatility governed by x_t .

Economic message: CIP deviations arise from the same risk-sharing pressure that drives UIP deviations. Real goods trade plus intermediary market power can create CIP wedges.

5.7 Table 2: simulated moments

Table 2
Simulated moments for frictionless and frictional models with taste shocks covariance $\rho^\Psi = -0.3$ and output-taste shocks covariance $s^\Psi = -0.4$.

	Data (S.E.)	Frictionless	Frictional
$ac1(de)$	-0.01 (0.09)	-0.00	-0.00
$std(de)(\%)$	9.41 (0.01)	13.38	11.33
$std(de)/std(dg_H)$	6.62 (—)	6.76	5.71
$std(d\tilde{c}_H)(\%)$	1.49 (0.10)	2.64	2.37
$std(d\tilde{c}_F)(\%)$	1.60 (0.11)	2.65	2.36
$std(de)/std(d\tilde{c}_H)$	6.32 (—)	4.21	3.99
$corr(d\tilde{c}_H, dg_H)$	0.79 (0.04)	0.65	0.72
$corr(d\tilde{c}_H, d\tilde{c}_F)$	0.61 (0.06)	0.04	0.06
$corr(d\tilde{c}_H - d\tilde{c}_F, de)$	-0.00 (0.10)	-0.68	-0.63
$std(r_H - r_F)(\%)$	0.69 (0.04)	0.00	0.14
$std(r_H - r_F)/std(de)$	0.07 (—)	0.00	0.01
$std(r_H)(\%)$	1.12 (0.07)	0.00	0.05
$corr(r_H, r_F)$	0.81 (0.03)	-0.96	0.74
$corr(dr_H, dr_F)$	0.59 (0.06)	-0.85	0.61
$ac1(r_H - r_F)$	0.96 (0.09)	0.99	0.95
$ac1(r_H)$	0.98 (0.09)	0.99	0.95
β^{Fama}	2.18 (1.25)	817.62	8.19
$Carry\ SR(\%)$	37.23 (18.41)	1.26	2.49
$Carry\ i\text{-}diff\ (\%)$	1.21 (0.39)	0.01	0.09
$Carry\ (\%)$	3.46 (9.28)	0.15	0.23
$Carry\ ratio\ (\%)$	34.97 (18.41)	0.06	0.66
$std(Carry)(\%)$	9.28 (0.60)	13.37	11.33
$std(Carry\ i\text{-}diff)(\%)$	0.39 (0.03)	0.00	0.04
$mean(CIP_H)(\%)$	-0.21 (0.30)	0.00	-0.17
β^{CIP}	-2.64 (0.68)	—	-0.37
t^{CIP}	-3.87 (—)	—	-2.36
$ERP_H(\%)$	3.48 (12.74)	0.36	0.36
$Equity\ SR_H(\%)$	27.26 (18.41)	4.47	4.48
$ERP_F(\%)$	2.01 (15.85)	0.34	0.34
$Equity\ SR_F(\%)$	12.66 (18.41)	4.23	4.23
$corr(ERP_H, ERP_F)$	0.85 (0.03)	0.35	0.35

The Data (S.E.) column presents observed values with standard errors in parentheses, enabling comparisons between frictionless and frictional models. For alternative parameter choices, see [Internet Appendix IA.1.2](#). Note that the empirical β^{CIP} reported in [Avdjiev et al. \(2019\)](#) is based on daily changes expressed in basis points, whereas the β^{CIP} in the simulated moments reflects monthly changes expressed in percentage points.

Figure 7: Table 2: simulated moments for frictionless and frictional models.

Read:

- Exchange-rate volatility is 9.41% in the data, 13.38% in the frictionless model, and 11.33% in the frictional model.
- The standard deviation of the interest-rate differential is 0.69% in the data, 0.00% in the frictionless model, and 0.14% in the frictional model.
- Cross-country interest-rate correlation is 0.81 in the data, -0.96 in the frictionless model, and 0.74 in the frictional model.
- The Fama beta is 2.18 in the data, 817.62 in the frictionless model, and 8.19 in the frictional model.
- Carry Sharpe ratio is 37.23% in the data, 1.26% in the frictionless model, and 2.49% in the frictional model.
- Average CIP_H is -0.21% in the data, 0.00% in the frictionless model, and -0.17% in the frictional model.
- The β^{CIP} coefficient is -2.64 in the data and -0.37 in the frictional model.

Economic message: The frictional model is not perfect, but it moves the key moments in the right direction: it creates interest-rate volatility, positive cross-country rate correlations, negative CIP deviations, and UIP/carry behavior that the frictionless model struggles to generate.

6 Short summary

- The paper builds a two-country continuous-time general-equilibrium model of exchange rates.
- Households can trade domestic assets directly but need intermediaries to access foreign financial claims.
- Intermediaries have access to a complete D2D market and bargain with households in D2C markets.
- Intermediaries are risk-neutral and can run matched books; the mechanism is market power, not balance-sheet capacity.
- Markups are state-contingent because households value some foreign claims more than others.
- These markups tilt household state prices and make international risk sharing inefficient.
- With frictions, monetary neutrality fails: money shocks can affect risk premia and exchange rates.
- The model derives closed-form expressions for risk premia, exchange rates, short rates, and CIP deviations.
- UIP deviations and CIP deviations come from the same underlying source: household demand for foreign risk-sharing filtered through intermediary markups.
- The calibration compares a frictionless model with a frictional model and shows that the frictional model can match several empirical patterns better.
- The frictional model generates negative average CIP deviations, positive rate correlations, realistic exchange-rate volatility, and more plausible UIP/carry behavior.
- The baseline model still struggles with equity premia and the full magnitude of carry returns, so extensions with asymmetric exposures and richer preferences may be needed.

7 Conclusion

7.1 Core conclusion

Malamud, Schrimpf, and Zhang show that intermediary market power can be a fundamental source of exchange-rate distortions. Even when global intermediaries are risk-neutral and maintain zero net positions, their ability to charge state-contingent markups in segmented D2C markets changes the state prices households face. This distorts risk sharing, breaks monetary neutrality, and produces UIP and CIP deviations.

7.2 Economic intuition

Households want foreign financial claims because they trade foreign goods and face country-specific shocks. But they do not have direct access to all foreign claims. Intermediaries provide that access and charge more in states where insurance is most valuable. These markups alter the household stochastic discount factor. Since exchange rates are ratios of state prices, the markups show up as exchange-rate risk premia, interest-rate movements, carry returns, and CIP deviations.

7.3 Takeaway

The paper reframes FX puzzles as consequences of imperfect intermediation. Rather than requiring intermediaries to hold risky inventories or hit binding capital constraints, the model shows that segmentation and bargaining power can be enough to create large wedges in international asset prices. The practical lesson is that exchange-rate dynamics depend not only on macro fundamentals but also on the market structure through which households obtain exposure to those fundamentals.